

Final Project – Mobile Systems (MobSys 2026)

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RH CALc – Relative Humidity Calculator

An Android application that calculates relative humidity from wet-bulb and dry-bulb temperature reading using the standard psychrometric method

1. Project Description

RH CALc is an Android application that calculates **Relative Humidity** from **wet-bulb, dry-bulb** temperatures using the standard psychrometric method. the user enters the two temperatures and depending the android device, if it has a barometer sensor it detects the pressure values and if not, it uses a constant pressure value of **1013.25** to compute for the **relative humidity**, present it as a formatted weather report with a graphical guage, and keeps a persistent history of past calculations. The project demonstrates essential Android development concepts required for MobSys 2026 which include; Activities, Intents, RecyclerView, CustomView, sensor integration and local persistence.

Platform:

- Andriod Studio (Iguana/2024.X or later)
- Kotlin
- minSdk 26, targetSdk 34 (minSdk kept low for device coverage; the app uses no APIs above 26 that lack a fallback)

2. Aim of the Project

To build a simple, correct, user-friendly Android app that calculates and displays relative humidity from wet-bulb and dry-bulb temperature readings, while satisfying all MobSys 2026 technical requirements.

3. Objectives

1. Develop an Android application using Kotlin.
2. Accept wet-bulb and dry-bulb temperatures input, with validation.
3. Calculate relative humidity based on user input.
4. Display results clearly across multiple Activities.
5. Transfer data between Activities using Intents.
6. Integrate two sensors and use a barometer (where present) as a real input to the calculation.
7. Provide a CustomView humidity gauge (Canvas).
8. Persist and display a calculation history using RecyclerView.

4. Application Design

Activity 1: Home Screen

Displays:

- App title
- Buttons: *Enter Temperatures*, *View Weather Report*, *History*
- Button: *History (RecyclerView)*
- Live sensor readings: *ambient light (Lux)* *shake detection*, *humidity sensor* and *current pressure (hPa, if barometer present)*

Activity 2: Temperature Input

User enters:

- Wet-bulb temperature (°C)
- Dry-bulb temperature (°C)
- Pressure (hPa) -auto-filled from a barometer or uses a constant pressure value of **1013.25**.

Buttons:

- *Calculate*, *Send to Report*

On calculate, the screen shows Tw, Td, pressure, and computed RH (%). Validated values are passed to Activity 3 via **Intent** extras.

Activity 3: Weather Report

Displays Tw, Td, pressure, RH, and date/time, plus the CustomView humidity gauge. Also derives and displays approximate altitude from the pressure reading (standard barometric formula) and the device has a TYPE_RELATIVE_HUMIDITY sensor, shows the device's own ambient humidity reading alongside the calculated RH as a validation check against the psychrometric formula. Receives data from Activity 2 and on display, writes the record to the Room database.

Activity 4: History Screen (RecyclerView)

Displays scrollable list of past records (Tw, Td, RH, timestamp) backed by **Room**, rendered via a custom RecyclerView.Adapter. Supports swipe-to-delete and a *Clear All* action.

5. CustomView - Humidity Gauge Description

A Canvas-drawn semicircular gauge:

- Semicircle arc with tick marks (0-100 %)
- Needle pointing to the current RH
- Colour zones: **Dry (< 30 %)**, **Normal (30-60 %)**, **Humid (> 60 %)**
- Smooth needle animation on load (ValueAnimator)

Implemented by extending View and overriding onDraw(Canvas); RH is set via a public setHumidity(value: Float) method.

6. Sensors Used

1. Light Sensor (TYPE_LIGHT)

- Reports ambient Lux; auto-switches light/dark theme

2. Accelerometer (TYPE_ACCELEROMETER)

- Shake gesture clear input fields and re-animates the gauge.

3. Barometer (*optional*) (TYPE_PRESSURE)

- Supplies real atmospheric pressure to the RH formula, falls back to manual field when absent. Also used to derive approximation altitude on the Weather Report screen.

4. Humidity (*optional*) (TYPE_RELATIVE_HUMIDITY)

- Reads the device's own ambient RH, shown beside the calculated RH on the Weather Report screen as a validation check against the psychrometric formula. Not used as a calculation input.

All sensors are accessed through SensorManager with registered listeners, unregistered in onPause() to save battery.

Note on the emulator: TYPE_PRESSURE and TYPE_RELATIVE_HUMIDITY may be unavailable on emulators and many phones. The manual pressure field and the “not available” fallback text guarantee the app is fully functional and demoable without either sensor.

7. Persistence

Local storage uses **Room** (SQLite), not SharedPreferences, because history is a growing list of structured records:

- @Entity CalculationRecord(id, wetBulb, dryBulb, pressure, rh, timestamp)
- @Dao with insert, getAll (ordered by timestamp desc), delete, clear
- Single @Database accessed via a repository

SharedPreferences is reserved only for the UI theme preference.

8. Technologies and Tools

Android Studio, Kotlin, XML layouts, Intents, RecyclerView + custom Adapter, CustomView(Canvas), Sensor Manager, Room, ViewModel/LiveData, and Android Emulator + physical device.

9. Methodology

1. **Planning:** Define features and UI flow.
2. **Design:** layouts for all Activities.
3. **Development:** Implement Activities, calculation module, sensors, gauge, and Room.
4. **Data Transfer:** Intents extras between screens.
5. **Testing:** Use Android Emulator and physical device.
6. **Documentation:** PDF write-up and screenshots.

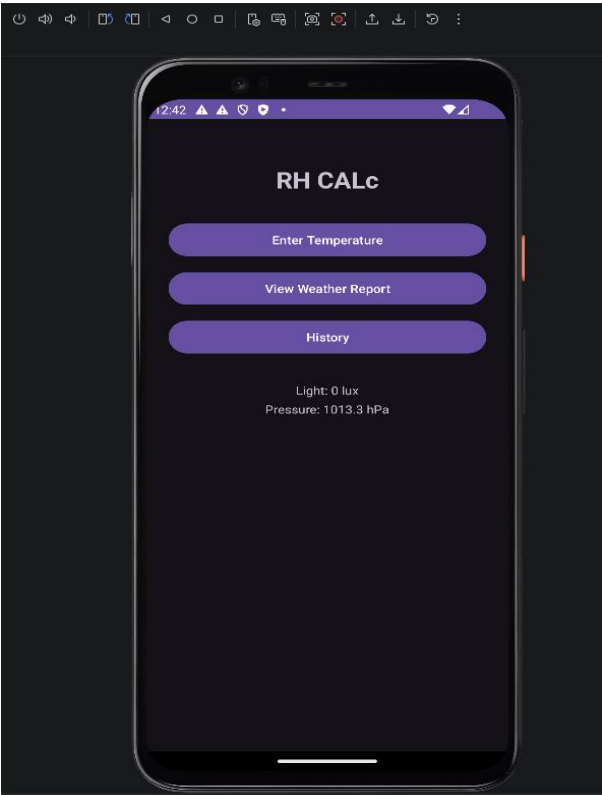
10. Expected Results

The finished app will accept wet/dry-bulb temperatures, compute correct relative humidity via the psychrometric formula, transfer data between Activities, render a graphical gauge, persist and display history via RecyclerView + Room, use the light and accelerometer sensors (plus optional barometer and humidity sensor inputs), validate all input, and present a clean UI.

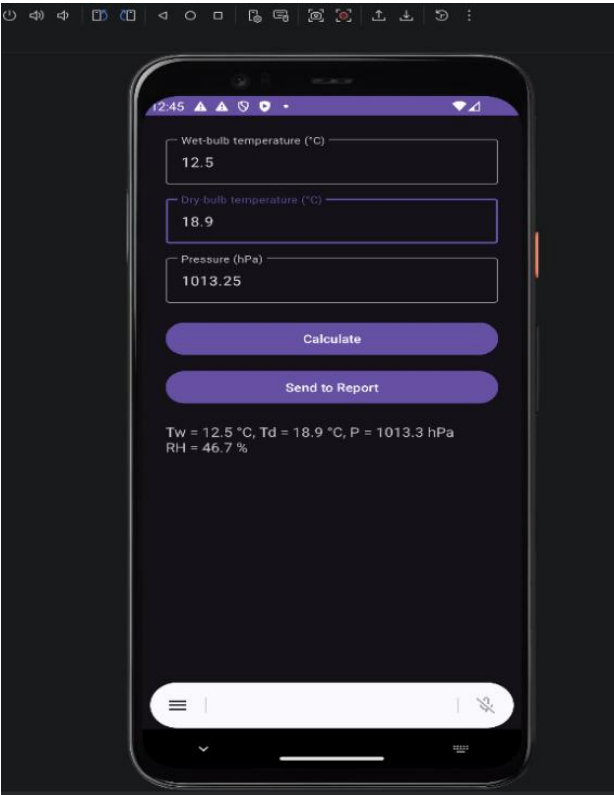
11. Screenshots (Placeholders)

(Replace with actual screenshots from your app)

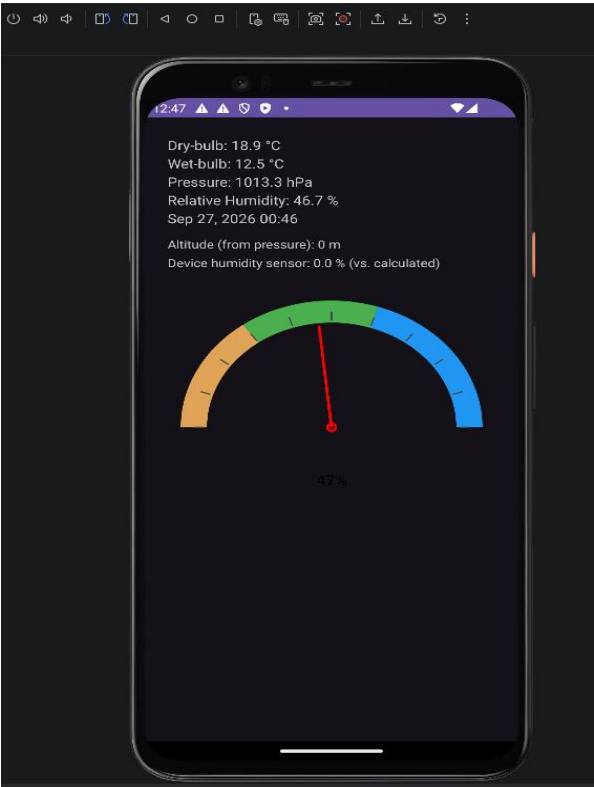
Home Screen



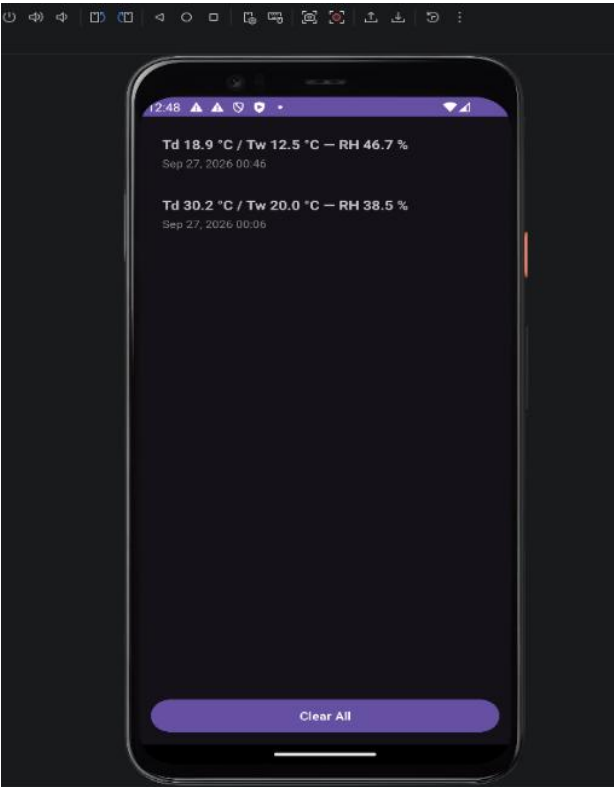
Temperature Input Screen



Weather Report Screen (with CustomView)



History Screen (RecyclerView)



12. Problems / Notes

- Barometer is absent on many devices and emulators – manual pressure fallback.
- Accelerometer sensitivity required a shake-detection threshold to avoid false triggers.
- Gauge animation needed Canvas optimisation (pre-allocated Paint / Path objects, no allocation inside onDraw).
- The psychrometer coefficient A assumes a ventilated instrument; a real reading depends on airflow over the wet bulb.

13. Conclusion

RH CALc is a complete Android application demonstrating the core mobile development concepts required for MobSys 2026; Activities, Intents, RecyclerView, CustomView, sensors, and Room persistence built around a correctly specified psychrometric humidity calculation with real pressure handling, input validation, and a clean, user-friendly interface.